Section A Comprehensive Review - Mixed

Directions: Show all work. Use exact values. Box or circle all answers.

1. Find a_7 if $a_1 = 2187$ and $r = \frac{1}{3}$.

2. A population has 45 organisms at the end of year 1 and doubles each year. State the common ratio and find the population at the end of year 5.

3. Find a_8 if $a_1 = 640$ and $r = \frac{1}{2}$.

4. A geometric sequence has $a_1 = -2$ and common ratio $r = -4$. Write the first five terms of the sequence.

5. Find a_5 if $a_1 = 4$ and $r = -2$.

6. Is 7,21,63,189, ... a geometric sequence? Explain.

7. Find the common ratio and write the next three terms: $5,10,20,40,\ldots$

8. Find a_6 if $a_1 = 3$ and $r = 4$.

9. Find the common ratio and write the next three terms: $4,12,36,108,\ldots$

10. A medication contains 1200 mg at the end of hour 1 and is cut in half each hour. State the common ratio and find the amount at the end of hour 4.

11. Find a_8 if $a_1 = 1$ and $r = 3$.

12. Is $4, -2, 1, -\frac{1}{2}, \ldots$ a geometric sequence? Explain.

13. Find the common ratio and write the next three terms: $1024,256,64,16,\ldots$

14. A population has 120 organisms at the end of year 1 and doubles each year. State the common ratio and find the population at the end of year 4.

15. Find a_6 if $a_1 = -2$ and $r = -3$.

16. A medication contains 640 mg at the end of hour 1 and is cut in half each hour. State the common ratio and find the amount at the end of hour 5.

17. Find the common ratio and write the next three terms: $625,125,25,5,\ldots$

18. Is $10,20,35,70,\ldots$ a geometric sequence? Explain.

19. A geometric sequence has $a_1 = -5$ and common ratio $r = 2$. Write the first five terms of the sequence.

20. Is $6,24,96,384,\ldots$ a geometric sequence? Explain.

21. Find a_7 if $a_1 = 320$ and $r = \frac{1}{2}$.

22. Is $28, -14,7, -\frac{7}{2},\ldots$ a geometric sequence? Explain.

23. Find a_6 if $a_1 = 972$ and $r = \frac{1}{3}$.

24. Is $3,6,12,25,\ldots$ a geometric sequence? Explain.